

Integrity & Basic Encryption



Chapter 6



Integrity & Basic Encryption

— — —

★ Definition

- Integrity, Trust, Sandbox

★ Hardware Integrity

- Physical Tamper Resistance

★ Basic Encryption

- Hash
- Symmetric Encryption
- Asymmetric Encryption
- Public Key

★ Security Protocol

★ Conclusion



Integrity

ความรู้ไม่พอ
แล้วใช้: เสี่ยง

ความรู้ Integrity คือสิ่งที่ขาดไม่ได้

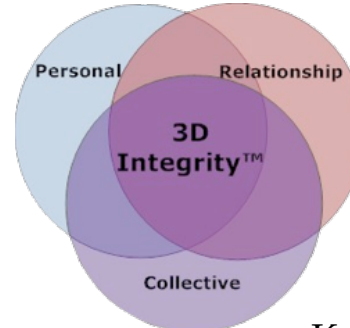
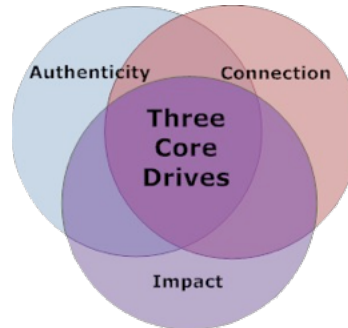
“Integrity without knowledge is weak and useless, and knowledge without integrity is dangerous and dreadful.”

Samuel Johnson (1709 - 1784)

- ★ Integrity is a characteristic that belongs to people who are self-actualized. It is the quality or condition of being whole, complete, unbroken, and undivided. Knowing oneself heightens a person's integrity.
- ★ Applying to data, it also refers to the state of being whole and undivided or the condition of being unified and unimpaired

combine

รวม





Authenticity: the forth A

- ★ Integrity is sometime referred as Authenticity--hence it is sometime mentioned as the forth “A” of security components.
- ★ How can we preserve the integrity of data?





Authenticity: the forth A

**Integrity requires
(hardware) supports.**

- ★ Integrity is sometime referred as Authenticity--hence it is sometime mentioned as the forth “A” of security components.
- ★ How can we preserve the integrity of data?





Trust

“Real integrity is doing the right thing, knowing that nobody's going to know whether you did it or not.”

Oprah Winfrey, in Good Housekeeping

★ Trust is a basis for every security model.

★ I trusted you. What does it mean?

- believe in the reliability, truth, ability, or strength of [M-W]

★ What do you trust?

- Trust Processor?
- Trust User/Developer?
- Trust Software/Process?



Trust-Based System in Action

— — —

- ★ **Single user** operating system with no protection
(DOS, CP/M, Windows < NT)
- ★ **Embedded** Systems/Appliances
(eg. Air conditioning controller)
- ★ **Old mobile phone**
- ★ What else?

minimal scales



How to minimize trust?

★ In reality, we cannot trust everything.

So, let's minimize trust.

★ Trust with respect to boundary

- Sandbox : isolated environment, run /w/o affect main system
- Domains (isolation)

↳ group comp, user, resource





Sandbox

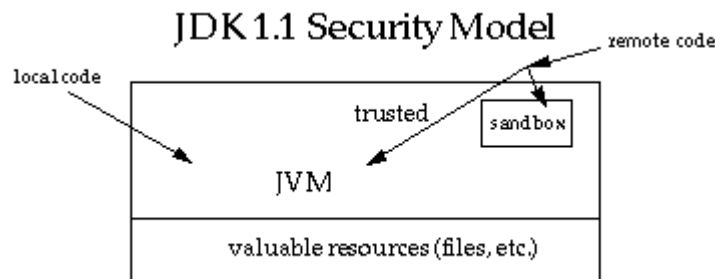
★ The sandbox security model is popularized by Java (Applet).

★ User has an option to trust an applet (remote code).

If trusted, the code is run with full local privilege.

If not trusted, the code will be in limited a sandbox.

(ie. cannot access local files.)





Sandbox (ctd.)

— — —

- ★ Trust decision is based on code signing.
 - ★ Who do you trust to sign your code?
 - ★ Is a signed applet (or ActiveX) a safe program?
-
- ★ Analogy. Is software that is developed by qualified developer (eg. Microsoft, Oracle) a secure software?
 - ★ Trust $\neq$ Secure



What mechanism
is necessary to
implement
sandbox and/or
domain?



Hardware Support for Integrity

L memory

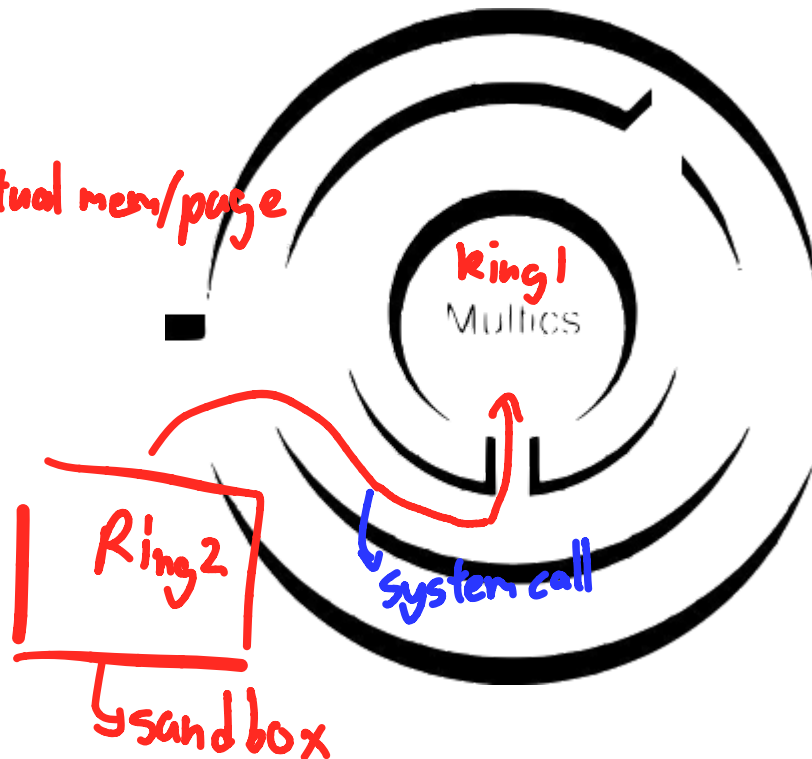
★ Electronic curtain

- Ring : position, memory mgmt / virtual mem/page

- Segmentation
- Page

★ Tagged Memory

So currently we write sandbox





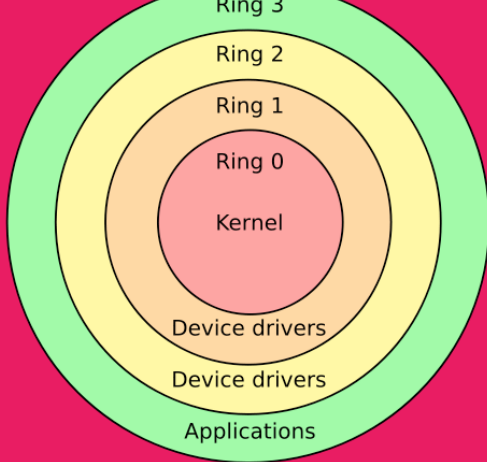


Ring

μ: micro
UNIX: simplified version Multics

- ★ Popularized by Multics (1960)
- ★ Processor provides memory compartments with each part associated with a ring.
 - Segmentation
 - Page
- ★ Inner ring (kernel) can access outer ring. Outer ring cannot only access inner ring though a specific method (ie. system call).
- ★ How many rings do we really need?

Actually 2 mostly used
But mostly defined 4 rings
1-2: device driver



Least privileged



user

admin

Most privileged

Fact

Intel x86 has 4 rings.

Most Operating Systems only use 2
rings (0,3)

★ IBM OS/2 is perhaps the only
operating system that deploy
device drivers in ring 1 and ring
2.



What if there is no hardware support?

Data in a portable drive, email, network packet do not always have hardware protection.



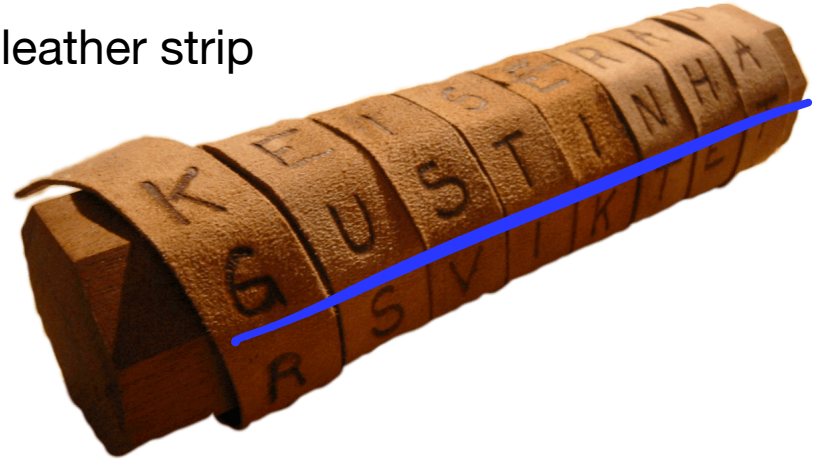
Confidentiality Integrity / Security
Confidentiality HW

Encryption as a tool

(Brief) History

Classical

- Greece (500 BCE)
 - Scytale - Spartan military encryption
 - Transposition cipher using rod and leather strip





History of Encryption

Key
plain text
cipher text

★ Monoalphabetic Substitution Cipher

★ Julius Caesar (50 BC) used it.

Caesar cipher

- abcdefghijklmnopqrstuvwxyz *normal*
 - SECURITYABDFGHJKLMNOPQVWXZ *non-sense*
- key: *normal key*

★ What is...? YRFFJ VJMFU
hello world

Lucas's key FPCSN



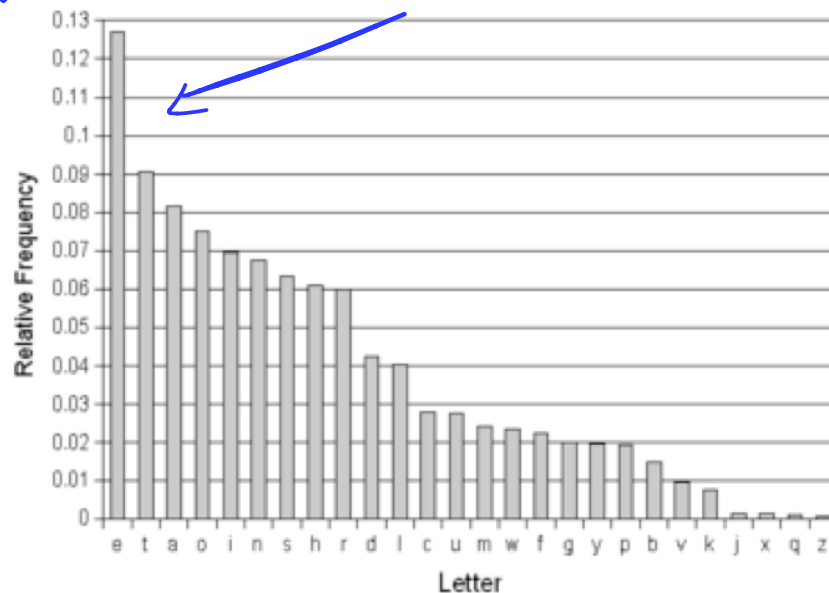
Pictures taken from https://www.amazon.com/DIY-Escape-Room-Bundle-Experience/dp/B08DJGM87T?ref=ast_sto_dp
<https://people.duke.edu/~ng46/collections/crypto-disk-strip-ciphers.htm>



Weakness of Encryption

- ★ Given a sufficiently large encoded message, it can readily be "cracked" by comparing the frequency of letter occurrences in the coded message with the frequency of letter occurrences in the language used for the message.

K ปรากฏบ่อย: e → ปรากฏบ่อย e





Try this

— — —

Gur fbyhgvba gb gur pvcure vf gur anzr nhoerl juvpu jnf qvssvphyg gb fbyir orpnhfr yvfgf bs cgbyrzi pbafrgryyngvbaf jrer vapbafvfgrag. Vg gbbx frireny snvyqr nggrzcgf orsber svaqvaf n jbexnoyr yvfg.



Encryption in Modern Day

— — —

★ Hash Digests (short input > long output)

- Message Digest (MD5, SHA-1, SHA-256, SHA-512, Tiger-Hash)

★ Symmetric Encryption

- Stream Cipher
 - eg. RC4 (used in SSL, WEP, WPA, etc...)
- Block Cipher
 - eg. AES, Blowfish, DES

★ Asymmetric Encryption

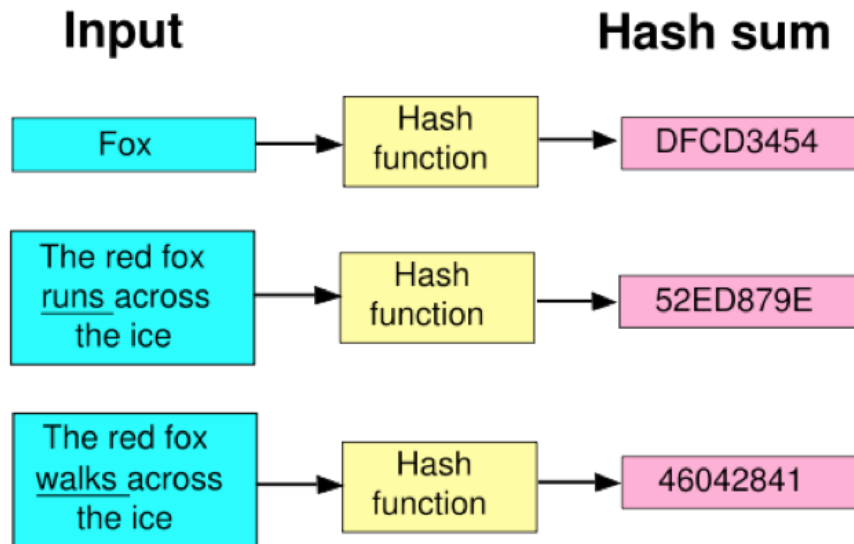
- Public Key
 - RSA
 - DSA



Hash (aka. Message Digest)

No key

- ★ Hash is a **one-way** function.
(Theoretically **no inverse function**)
- ★ Input may vary in size, **output** is a **fix-length value**.
- ★ Collision?





Food for Thought: Hash and Integrity

- ★ Hash Value is used widely for integrity check.
(Eg. To validate that software downloaded from mirror sites is the same as the original software.)
SW (software) function 2 ilu (2 lines)
1-way func
- ★ Since it cannot be decrypted, hash is sometime not considered an encryption.



[SHA512SUMS](#)



[SHA512SUMS.sign](#)



[debian-9.9.0-amd64-netinst.iso](#)



[debian-9.9.0-amd64-xfce-CD-1.iso](#)



[debian-mac-9.9.0-amd64-netinst.iso](#)



Stream Cipher

- ★ Vigenere (1467) is among the first stream cipher.
- ★ Extend the key to match the size of input. *based on algorithm*
↳ How to extend
- ★ Depending on a position, a same letter may encrypt differently.

original key

LUCKLUCK ---- Key (extended)
COMPUTING ---- Plain Text
NIOZSECPQ ---- Cipher Text

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
2	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
3	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
4	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
5	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
6	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
7	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
8	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
9	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
10	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
11	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
12	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
13	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
14	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
15	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
16	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
17	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
18	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
19	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
20	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
21	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
22	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
23	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
24	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
25	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
26	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z



Block Cipher

one input block, input : block
output : block

- ★ Wheatstone-Playfair Cipher (1854) : 2 chars for 1 block
- ★ Use heavily during the World War II
- ★ Construction 5x5 cipher blocks.
- ★ Grouping data in two, and use the corner to encode data.
- ★ Depending on a group, a same letter may encrypt differently.

F	I	R	S	T
A	M	E	N	D
B	C	G	H	K
L	O	P	Q	U
V	W	X	Y	Z

same col
shift down
same row
shift right
All diff
op corner

key : First Amendment

Example,

C	O	M	P	U	T	E	R
O	W	E	O	Z	D	G	E

For more details, see https://en.wikipedia.org/wiki/Playfair_cipher



Mode of Operation in Block Cipher

★ Variations

- Initial Vector
- Padding : 72 bit block, 1000:15000
- Chaining
- Feedback

★ Some modes may still be easier to find patterns.

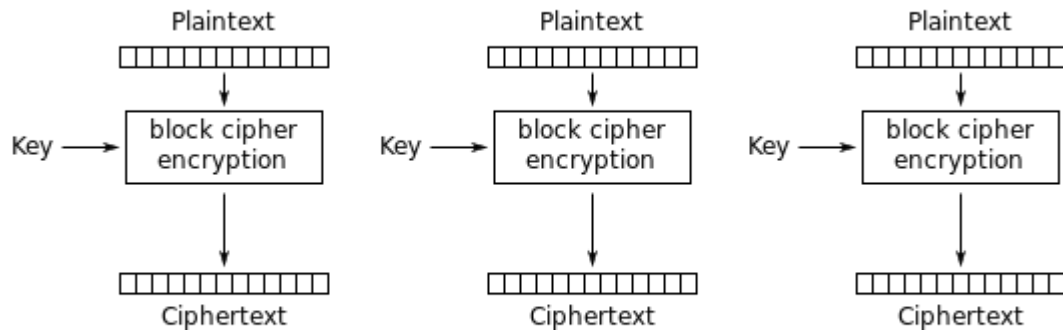
- ★ Electronic codebook (ECB)
- ★ Cipher block chaining (CBC)
- ★ Propagating CBC (PCBC)
- ★ Cipher feedback (CFB)
- ★ Output feedback (OFB)
- ★ Counter(CTR)
- ★ Galois/Counter (GCM)



Electronic codebook (ECB)

Block to Block without feedback or IV

Pattern is still there?



Electronic Codebook (ECB) mode encryption

Original

CHULA ENGINEERING
Foundation toward Innovation

COMPUTER

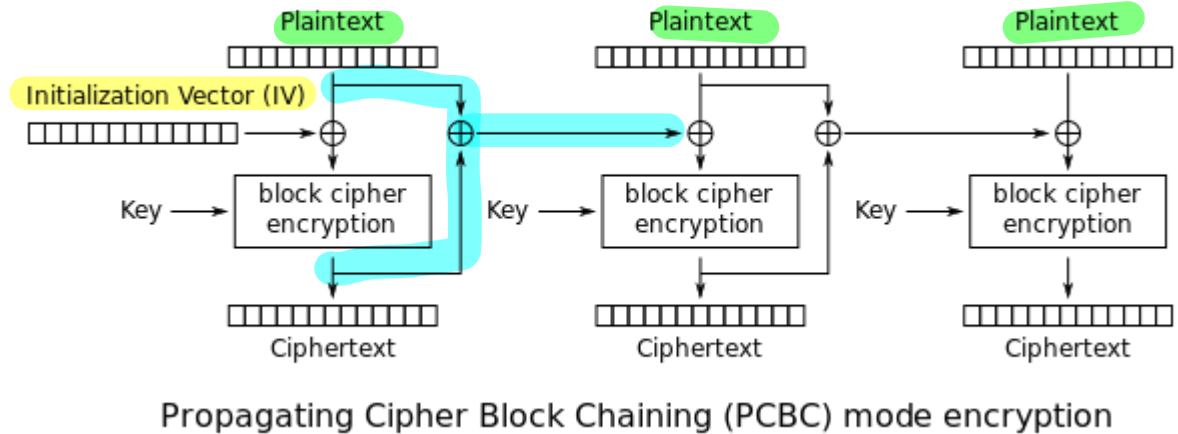
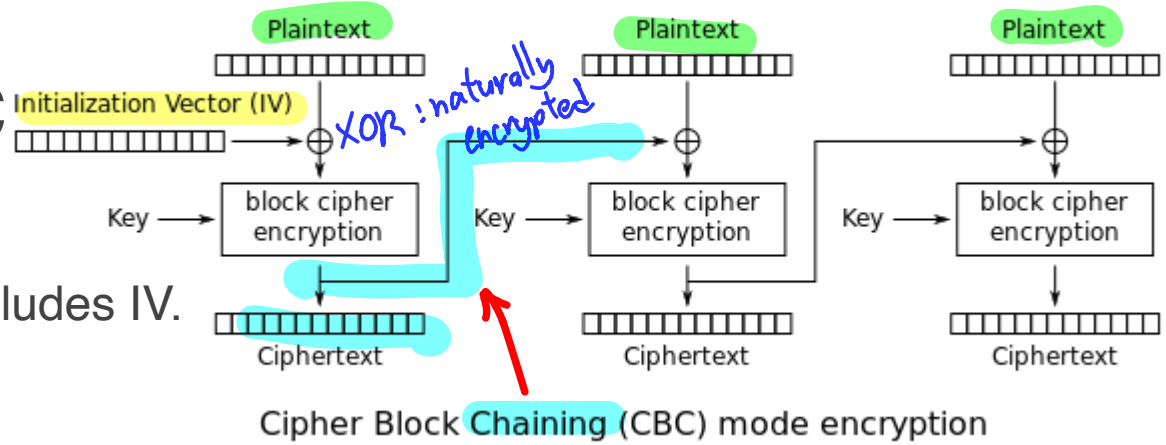
aes-256-ecb





CBC & PCBC

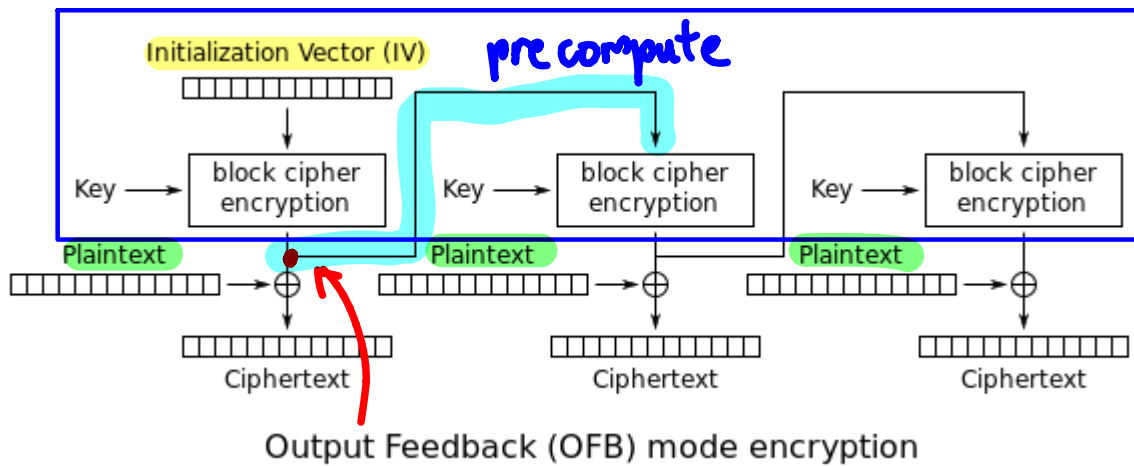
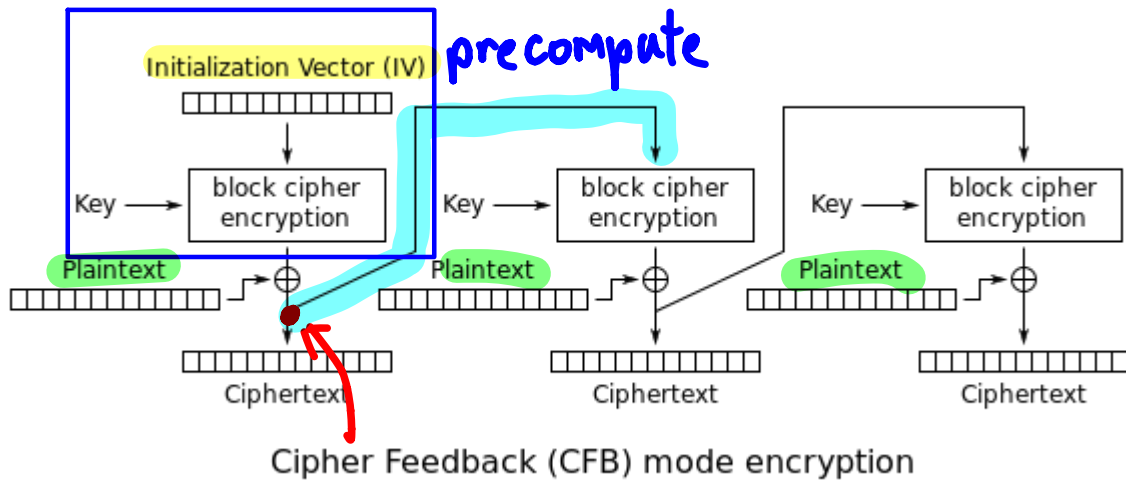
★ CBC (and variations) includes IV.





CFB & OFB

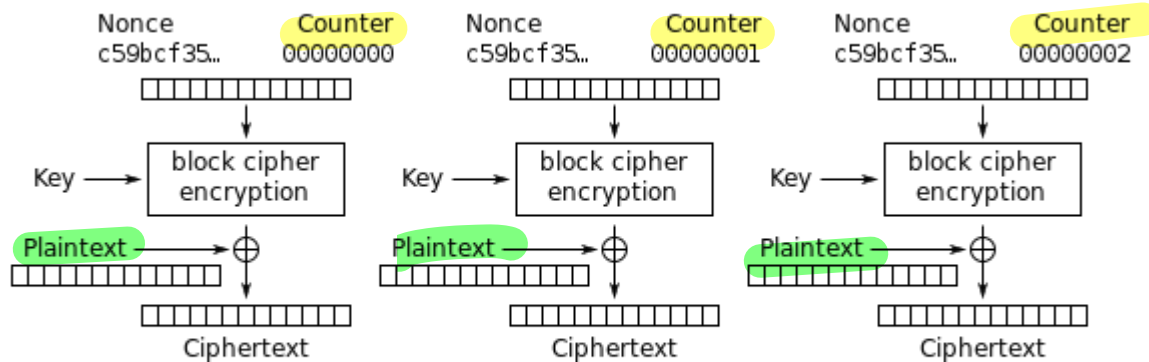
- ★ CFC (and variations) includes IV and use output as a feed for next block.





Counter

- ★ Counter (and variations) use counter and nonce as a vector for each block.



Counter (CTR) mode encryption



Food for Thought: Scalability of Symmetric Encryption

— — —

- ★ Assuming that a professor wants to share a piece of information with 100 students, how many (symmetric) key do we need in order to prove the integrity of the information? (ie. proof that the document is created by a professor and only the professor.)

150

so its not scales

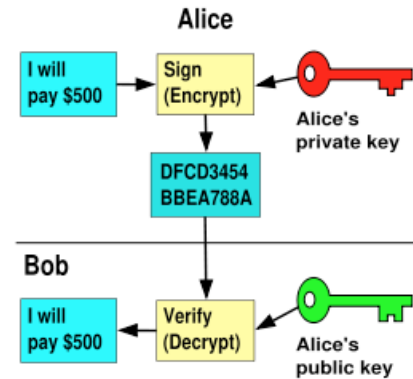
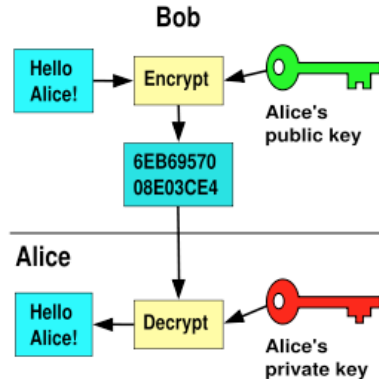
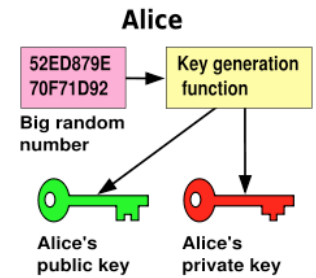
- ★ Hint.

With one key, anyone (with the key) can write a message.



Public Key

- ★ Based on prime numbers, a key pair (private key and public key) is created.
- ★ Message encrypted with a private key can only be decrypted with the associated public key--vice versa.





Public Key (ctd.)

— — —

- ★ Now, we **only have to keep the private key**. Our public key can freely be distributed. (eg. posted on our personal page.)
- ★ Note that there is still an issue with the physical identity of the public key owner.



Food for Thought: Public Key

★ Bob wants to send a message to Alice in a way that only Alice can read and Alice can prove that the message is coming from Bob. How?

(Which one is more secure?)

- ~~A~~ $((\text{message})_{\text{Bob's Priv}})_{\text{Alice's Pub}}$
- B. $((\text{message})_{\text{Alice's Pub}})_{\text{Bob's Priv}}$

wn: outer dont know
Alice's Pri
so they cant
break first box

Modern Cryptography Challenges

- Threats

- Quantum Computing
- Machine Learning Attacks
- Side-Channel Attacks
- Computational Power Increase

- Research

- Post-Quantum Cryptography
- Blockchain Technologies ?



Fact

Public Key is slow.
Block/Stream Cipher is faster.
Hash function is the fastest.

- ★ Algorithmically, Public Key is based on exponential functions. It is slow. Stream Cipher and Block Cipher are based on logic function. It is faster.

— — —



Fact

Encryption for Authentication and Authorization.

- ★ With proper key management, encryption can provide both authentication and authorization.
- ★ For authorization, it can provide both confidentiality and integrity.
- ★ However, encryptions are vulnerable (to statistical analysis). With enough processing power, everything can be cracked.

— — —



Security Protocol

— — —

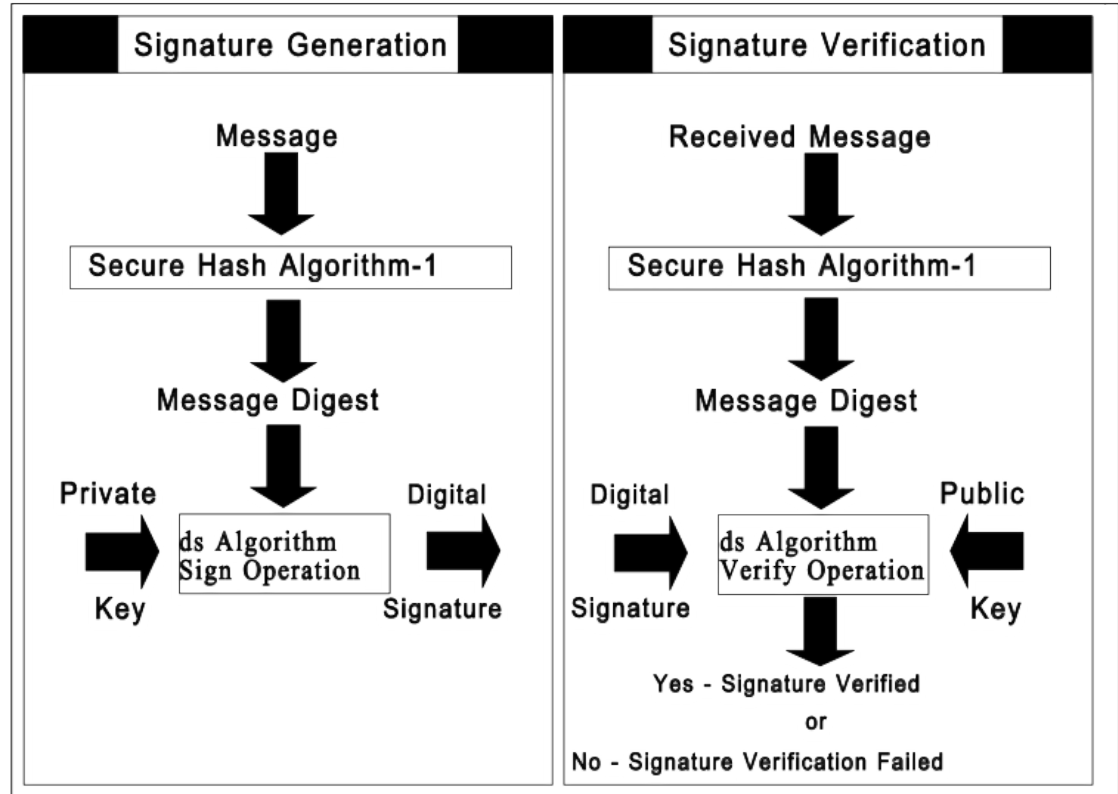
- ★ Security Protocol is usually a combination of several encryption algorithms. They usually combine pros and cons of several algorithms.



Public-Private key

Security Protocol: Digital Signature

- ★ A receiver can verify the originality of the a (plain) text.
- ★ Combine the speed of message digest with the scalability of public key.





Security Protocol: HTTPS/SSL/TLS

★ Combine speed of symmetric key with scalability of public key.

Client

Server

Depending on the environment, client may select stream/block cipher algorithm with a key.

1) $\text{https://www.chula.ac.th/}$

2) Here is my Pub & certificate.

3) Let's use RC4 with this (key)

Server's Pub

encrypted (key)

Exchange (Data)_{key}

Since only server has a private key, only client and server share the same **key**.

key exchange

data exchange

SSL/TLS Protocol

นี่คือขั้นตอน

Note: We will revisit the certificate later.



Pop Quiz

★ What kind of security do we get from encryption?

Authen, Integrity (needs key too)

★ Most people think that encryption is secure.

Do you agree or disagree? Not always



Conclusion

— — —

- ★ Integrity requires either hardware support or encryption.
- ★ A combination of hardware and Operating System serves as a basis for integrity (security).
- ★ A box with strong integrity means it will always keep things inside in the original conditional for years.
- ★ There is not absolute integrity. Most of the time, a security breach is the break of integrity. (eg. A thief broke a window and got into a property.)

Conclusion

- Key Takeaways

- Cryptography is a dynamic, evolving field
- Critical for modern digital security
- Requires continuous learning
- Blend of mathematics, computer science, and security

- Further Learning

- Understand fundamental principles
- Practice implementation
- Stay updated with emerging technologies
- Develop critical analytical skills



Conclusion (ctd.)

— — —

★ Contemporary Algorithms

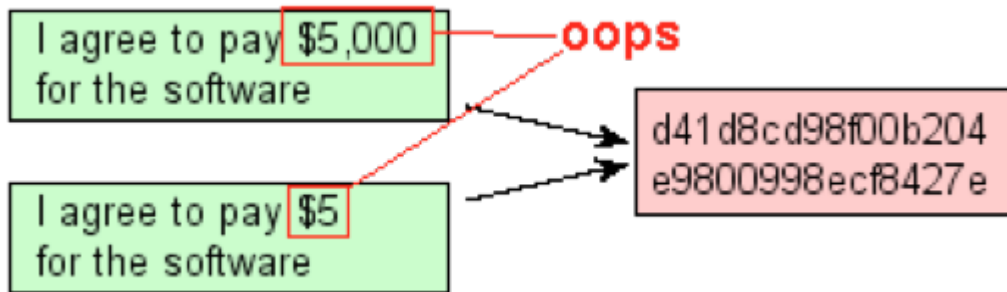
- Hash/Digest Algorithms
 - ~~MD5, SHA-1~~, SHA-256, SHA-512
- Stream Ciphers
 - RC4, CHACHA, SEAL
- Block Ciphers
 - Blowfish, DES, 3DES, AES
- Public Key
 - RSA, DSA



Conclusion (ctd.)

★ Other issues not in this chapter:

- Key management (We will revisit this in PKI later.)
- Collision of hashing function <http://www.mathstat.dal.ca/~selinger/md5collision/>
- Post-Quantum Encryption
- Elliptic-curve cryptography





End of Chapter 6